

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients**, processing **10,000+ monthly submissions** via Mayden iaptus REST data integrations; contributed to **\$340K+** in revenue.
- Architected a full-stack ML training-data pipeline and annotation platform (React, Node.js, MongoDB) used daily by the AI team, cutting manual data work by **100+ hours/month** and lifting labeling throughput by **60%**.
- Implemented data-governance controls, **AWS KMS** encryption for sensitive PII/clinical data, 90-day retention, auth hardening, and rate limiting, and remediated **20+ VAPT findings**.
- Served as primary technical POC, communicating results to stakeholders, mentoring a second developer to on-call readiness, and piloting a COO-sponsored GitHub Copilot AI-assisted-development experiment.

Cario Growth Services, Backend Developer Intern | *Chennai, India* Nov 2022 – May 2023

- Designed and shipped **20+ RESTful APIs** (Node.js, Fastify, **PostgreSQL**) with input validation, clear contracts, and **SQL** querying used in production by the frontend team.
- Migrated core repositories to TypeScript and shipped a production LLM automation feature (auto-commenting via **4-bit quantized Llama-7B**, **llama.cpp**) with async queued generation and content-safety filtering.

Freelance / Open Source, Full-Stack Engineer | *Remote* Jul 2022 – Aug 2023

- Built and deployed **Underdogs Fitness** (Website), a production MERN platform with Stripe hybrid payments, cron automation, and a multi-branch data model; live **3+ years**.

Projects

Spark ETL Performance Optimization (GitHub) PySpark, Spark UI, SQL, NYC TLC Dataset

- Built and tuned a **PySpark** ETL pipeline on a **~1.4B-row** dataset, extracting, transforming, and loading trip data with broadcast joins, adaptive execution, skew-join handling, partitioning, and column pruning to cut end-to-end runtime **~25%** (17–19 min to 14.3 min) and tail-latency ratio **2.14x to 1.74x**, profiled via Spark UI.

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a Python agent that automates open-ended queries into structured entity tables via typed retrieval planning, reranking, and deterministic extraction with LLM fallback and **cell-level provenance**, an intelligent data-processing workflow hardened with **150+ tests**.

KG2RAG-Enhanced – Multi-Hop QA on HotpotQA (GitHub) Python, Ollama / LLaMA-3, sentence-transformers, RRF

- Built the multi-view retrieval component of an ML RAG pipeline (query-side decomposition, RRF fusion, cross-encoder reranking, MMR), evaluated on a **4,905-question** set with 95% binomial confidence intervals, lifting supporting-fact recall **+2.68%** over the KG²RAG baseline.

Publications

- “Alzheimer’s Disease Classification using Transfer Learning,” IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: Python, SQL, JavaScript, TypeScript, PostgreSQL, MongoDB, SQLite

ML & Analytics: Pandas, NumPy, scikit-learn, PyTorch, statistics, regression / classification, evaluation

Data & Pipelines: PySpark, Spark, ETL, data pipelines, Spark UI, Hadoop / MapReduce

AI & Agents: LLM agents, RAG, agentic pipelines, OpenAI / Groq APIs, GitHub Copilot

Cloud & DevOps: Azure, AWS, GCP, Git, GitHub Actions, Docker, Linux

Achievements

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.